Marching Squares

Coarse saddle fixtures, measured continuity and reconstruction-safe geometry

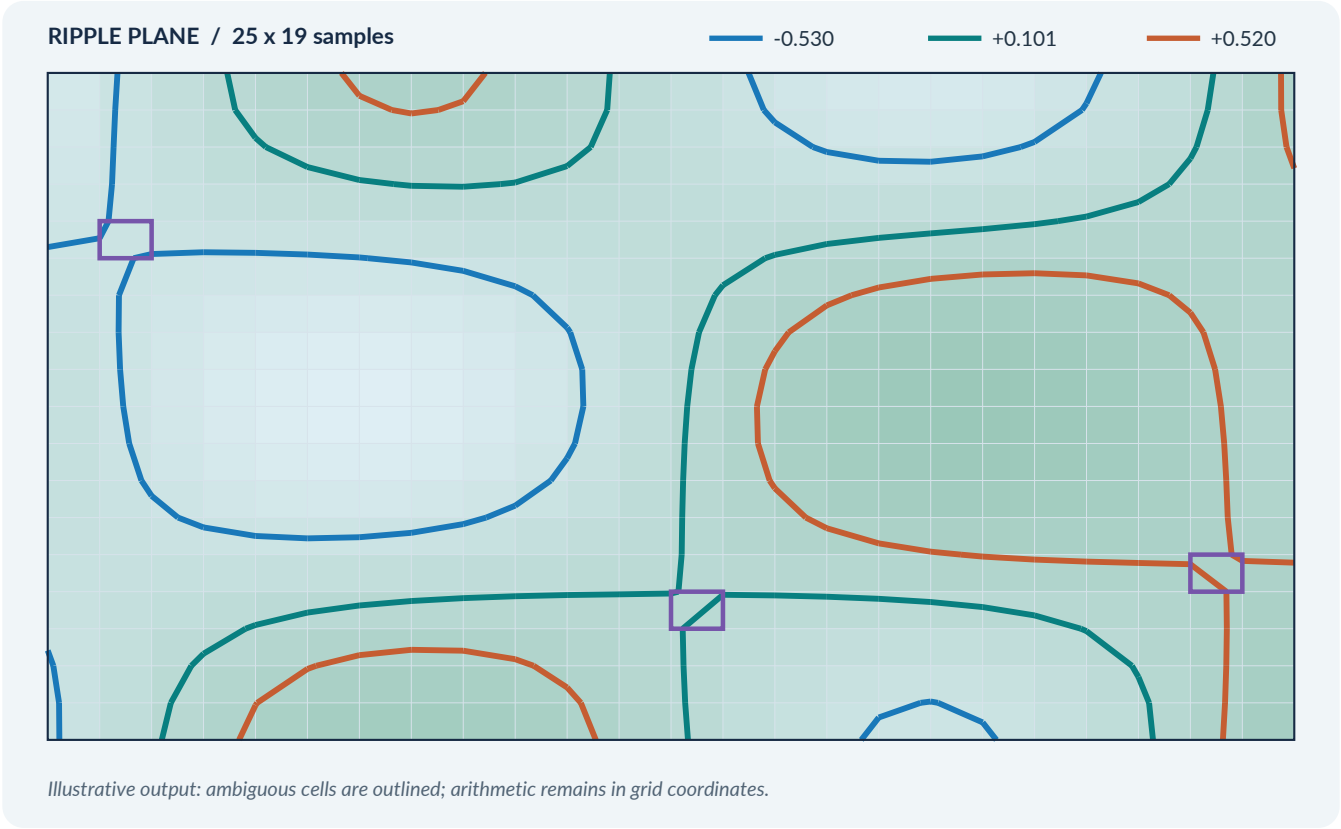


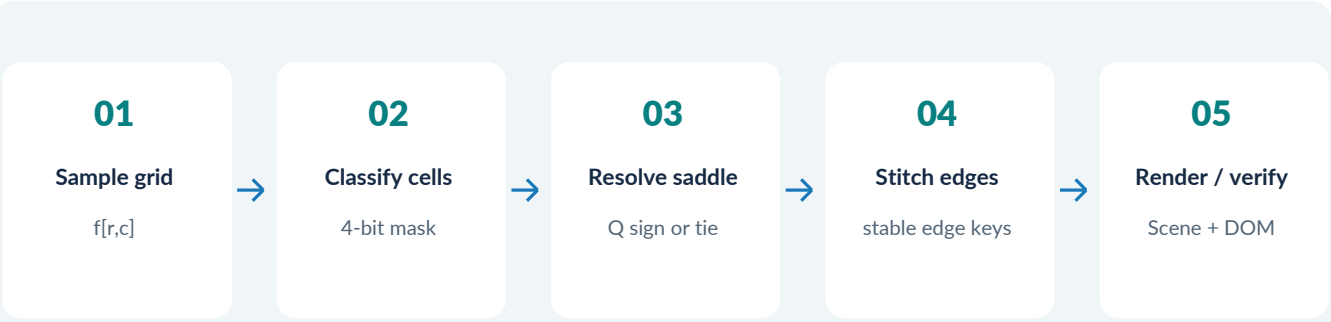
Figure 1. Default Ripple field at three ambiguity-bearing levels. Purple rectangles mark cases 5/10; exact pairing and visibility are checked independently of the artwork.

Document	Target	Revision	Status
Design specification	svg-dom-graph demo-app	V3.0 · 4 Nov 2026	Approved for Phase 0

Primary proof obligation A contour is a world-coordinate topological result. Every adjacent cell agrees on a shared grid-edge crossing, and neither the scene view nor repeated scene construction may change its connectivity.

01 Outcome and integration contract

The demo extracts isolines from a sampled 2D scalar field at three preset threshold levels, including cases 5 and 10 in the default Ripple and a coarse field that exposes multiple saddles at 1×. Its purpose is to expose failures in path construction, plotting, zoom, DOM replacement, event lifecycle and accessibility—not to add contour computation to the library itself.



Each stage consumes world-coordinate data. Zoom never enters the solver.

Requirements

- **R1 - Real algorithm:** each cell has a 4-bit case; interpolation uses scalar samples, and ambiguous cases take the bilinear decider.
- **R2 - Multiple levels:** show three thresholds together or isolate one; levels have unique labels and distinct line patterns as well as colours.
- **R3 - Inspectable topology:** report the selected cell, four samples, case index, crossed edges, Q, normalized diagnostic rho, segment pairing and open/closed totals.
- **R4 - Deterministic stepping:** Previous, Next, Restart and seek yield the same projection regardless of navigation history.
- **R5 - View invariance:** pan/zoom changes only view transforms; solver edge keys, coordinates and graph topology stay unchanged.
- **R6 - Rebuild invariance:** field/level/step/selection/view survive deliberate reconstruction without new gaps, duplicate listeners or stale DOM.

Boundary of responsibility

The **demo-app** owns the scalar samples, Marching Squares solver, contour stitching, inspector and controller. **svg-dom-graph** owns the *Scene*, plot-node rendering, zoom/pan and controls. No new public crate API is required by default. The stable outer stage SVG owns controls outside the zoomed Scene content. Phase 0 verifies PlotLine, PlotPoints, clipping and scene layering; a demo-local closed-path overlay is permitted only *inside* the transformed content group.

Verified existing API/documentation: `Scene::add_plot_node`, `set_plot_items`, `set_plot_description`, `on_plot_click`, `view`/`set_view`, *zoom toolbar and* `svg_buttons`. *Source reference: main branch* `Cargo.toml` *v0.2.47 and repository documentation accessed 4 Nov 2026.*

02 Demonstration and user interaction

Composition and layering

A stable outer <svg> holds the zoomed Scene content as a replaceable subtree. Buttons and the range are library SVG controls in a fixed, unzoomed stage layer; native <select> and <input> controls live in <foreignObject> there. The external live region and inspector controls survive swaps. Plot data occupy grid x = 0..W−1, y = 0..H−1, with row 0 at y = H−1; rendering axes add 0.25 grid units of padding on each side. Three distinctive contour line patterns appear over light sample markers. The inspected threshold is the middle slot by default; at k=0 there is no default selected cell.

Control	Action / invariant
Field selector	Ripple plane (default), Twin hills, Coarse saddle waves, Saddle lab. A switch selects final step C. Preserve view only for matching dimensions AND extents.
Levels: All / one	Three contours or one; the low / middle / high style and legend identity belong to slots, never to raw numeric values. The inspected-level selector is independent.
Threshold edit	Native numeric input (step 0.001): require finite, strictly increasing low < middle < high, and no sample-on-level. Reject/revert invalid edits. Slot styles never reorder.
Prev / Next / Restart / seek	Snapshots k=0..C are projections of immutable cell records; no history stack or animation playback.
Inspector: row + column	Keyboard-operable native selectors; plot click selects exactly the same cell through the zoom transform.
Rebuild scene	Replace Scene content only. Preserve step, level mode, selection and focus; retain view when compatible.
Zoom toolbar	Use the existing zoom/pan implementation; do not regenerate numerical geometry.

Prefix projection and inspection

Snapshot 0 shows sample markers, but no contour segments. Snapshot k uses exactly the cells with row-major indices < k, at every active level. Run the **same stitcher** on that prefix. Degree-1 endpoints along the processed/unprocessed frontier are provisional open ends; a cycle entirely inside the prefix remains a genuine closed loop. Never fabricate an edge across unvisited cells, and do not label an open partial chain as a closed loop.

At snapshot **k = 0**, no cell is selected by default: the inspector reads “No cell selected”. The inspector otherwise follows the last processed cell unless explicitly changed. For any **unvisited** selected cell it shows coordinates and “Not yet evaluated”; never reveal mask, Q, rho, intersections or pairing. At completion it reports open, closed and exact-saddle totals per level.

Saddle lab and accessibility

The 2×2 lab presents five named test matrices at threshold zero, including both Q signs for each ambiguous mask and a singular Q=0 crossing. The additional 9×7 coarse preset is a production visual reference with easily inspected, nonzero-Q saddles. The very same solver produces all paths. Controls remain a constant visual size at 0.25× zoom, and all inspection is keyboard-accessible. An external polite live region speaks manual changes without repeating on rebuild. A textual legend and line dashes distinguish levels without relying on colour alone.

03 Data, fixtures and numerical contract

Core types (illustrative Rust, not new crate API)

```
struct Grid { width: usize, height: usize, values: Vec<f64> }
struct LevelTrace { level: f64, by_cell: Vec<CellResult> }
struct CellResult { mask: u8, q: Option<f64>, segments: Vec<Segment> }
enum VertexKey { H { row: u16, col: u16 },
                 V { row: u16, col: u16 },
                 SaddleV { row: u16, col: u16 },
                 SaddleH { row: u16, col: u16 } }
struct Segment { a: Crossing, b: Crossing }
struct Crossing { key: VertexKey, point: Point2 }
struct Contour { vertices: Vec<Point2>, closed: bool }
```

Sampling and configured levels

Preset	Sampling	Scalar function / thresholds
Twin hills (comparison)	25 × 19	$1.55\cdot\exp(-((x+1.25)^2/1.15 + (y+0.1)^2/0.92)) + 1.35\cdot\exp(-((x-1.1)^2/1.0 + (y-0.25)^2/1.05)) - 0.28$; x in [−3,3], y in [−2.25,2.25]; levels −0.12, +0.24, +0.66
Ripple plane (default)	25 × 19	$\sin(1.15x)\cdot\cos(1.35y) + 0.16x - 0.08y$; same domain; levels −0.53, +0.101, +0.52
Coarse saddle waves	9 × 7	$\sin(x)\cdot\sin(y)$, x = c−3.5, y = 2.5−r, for integer c = 0..8 and r = 0..6; levels −0.10, +0.10, +0.20.
Saddle lab	2 × 2	Explicit corner matrices, one cell; levels −0.5, 0, +0.5; the named case fixtures are evaluated at level zero.

Validation and limits

- Require 2 ≤ width ≤ 65, 2 ≤ height ≤ 65, width × height ≤ 4,225, exactly width × height finite samples and exactly three finite strictly ascending levels (low < middle < high). Reject invalid input before any DOM mutation.
- All preset levels and accepted edits must have no grid-vertex sample equal to any configured threshold. Solver returns a typed `VertexOnLevel` error for equality; an entire grid edge at level is not silently omitted or converted into a fabricated isoline.
- Empty levels are legitimate. Limit fields to 4,096 cells × 3 levels; reject exceeded segment, endpoint-incidence, stitched-path or render-item budgets with a typed `GeometryBudgetExceeded`. Use checked multiplication before allocating.
- All arithmetic is f64 in model grid coordinates. No rounding, screen pixels, zoom scale or CSS dimensions enter classification, intersection calculation or stitching.

Exact sample-on-level equality remains unsupported in V3. In particular the initially proposed Ripple +0.100 equals the sample at x=0, y=−1.25; substitute +0.101 (nearest sample distance 0.001). Production native tests assert the precise Ripple mask/Q sign fixtures, the coarse counts (6/6/2 ambiguous cells), and no sample within 1e−4 of any preset level. A separate perturbation policy is out of scope.

03A Visibility, saddle counts and normalized diagnostics

The three Ripple levels each exercise a genuinely ambiguous cell. Their Q values are small in absolute magnitude, but that is not a reliable measure of drawn-contour separation. The exact, straight segments generated by Marching Squares have the minimum separations below. These are **model-space** checks, not a guarantee about browser stroke joins or line visibility. Geometry must still be inspected at 1× and 4×.

Ripple level	Case / (row,col)	Q	rho	Segment gap
−0.530	10 / (4,1)	-7.889×10^{-4}	0.892	0.848 grid
+0.101	10 / (14,12)	-4.827×10^{-5}	0.425	0.708 grid
+0.520	5 / (13,22)	$+1.650\times10^{-4}$	0.416	0.757 grid

The minimum gap is the shortest Euclidean distance between the two disjoint *piecewise-linear* Marching Squares segments in that cell, not the distance between continuous bilinear isolines. At the nominal 25-pixels-per-grid-unit layout the three gaps are approximately 21.2, 17.7 and 18.9 pixels. All three production cells have rho > 0.10 and therefore **do not** trigger the proposed near-degenerate diagnostic.

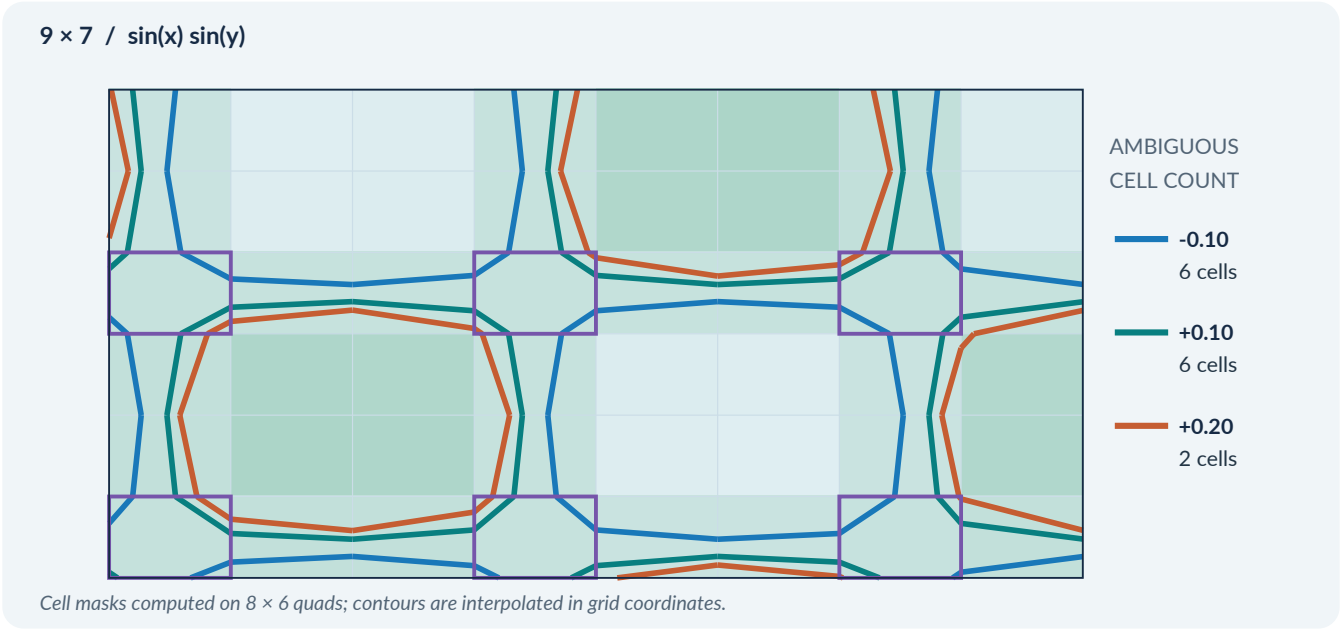


Figure 2. Coarse saddle waves at −0.10 / +0.10 / +0.20. Purple boxes identify ambiguous cells from all three iso-levels. The plot is a schematic generated from the same segment rules, not a browser-API guarantee.

Coarse preset: exact acceptance fixtures

Use **9×7 samples** f(c,r) = sin(c−3.5) sin(2.5−r), for c = 0..8 and r = 0..6. Its three ascending levels −0.10, +0.10, +0.20 contain respectively **6, 6 and 2** ambiguous cells. Both mask 5 and mask 10 occur at each of the first two levels; the final level contains mask 10. In this preset every such Q is nonzero. Native tests must assert these counts, Q signs and the solver's actual pairings without an exact-Q tie assumption.

Threshold perturbation oracle

For Ripple cell (14,12), +0.101 yields case 10, Q ≈ -4.827×10^{-5} and the T–L / R–B pairing. At +0.102 it remains case 10, but Q ≈ $+4.312\times10^{-5}$ and switches to T–R / B–L. The move is **intentionally topological**, not a no-op; restoring +0.101 must restore the original canonical digest.

04 Cell cases and edge interpolation

Order the four corners **TL, TR, BR, BL** and use `g = sample − threshold`. Set a bit when g > 0: TL = 1, TR = 2, BR = 4, BL = 8. Edges are T (top), R (right), B (bottom) and L (left). For any crossing with endpoints gA, gB of opposite signs: **t = gA/(gA−gB)**, and **p = A + t·(B−A)**. The same edge is evaluated once and shared between both incident cells.

Mask	Active corners	Edge pairing
0 / 15	none / all	none
1	TL	T–L
2	TR	T–R
3	TL+TR	L–R
4	BR	R–B
5	TL+BR	ambiguous; decide by Q
6	TR+BR	T–B
7	TL+TR+BR	L–B
8	BL	B–L
9	TL+BL	T–B
10	TR+BL	ambiguous; decide by Q
11	TL+TR+BL	R–B
12	BR+BL	L–R
13	TL+BR+BL	T–R
14	TR+BR+BL	T–L

Coordinate conventions

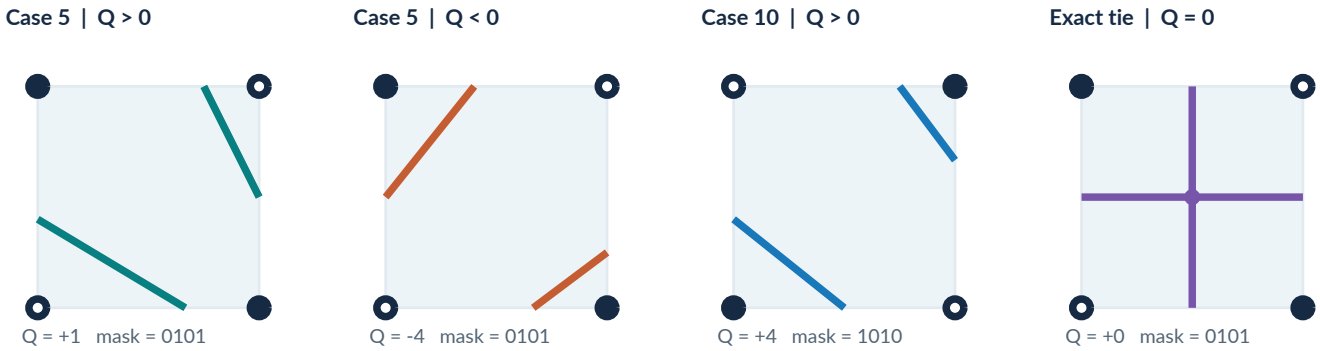
Cell (r,c) has TL=(c, H−1−r), TR=(c+1, H−1−r), BR=(c+1, H−2−r) and BL=(c, H−2−r). Plot coordinates are those same coordinates; do not invert y again while drawing. Edges use canonical keys: T→H(r,c), B→H(r+1,c), L→V(r,c), R→V(r,c+1). Every crossing key owns one interpolation result; adjacent cells reference it unchanged.

Determinism

Visit cells in row-major order, levels in displayed order, and emit pairs in the table order. Define VertexKey ordering as lexicographic (kind,row,column), where H < V < SaddleV < SaddleH. Horizontal edge interpolation is always left-to-right; vertical interpolation is always top-to-bottom. A contour starts from the lowest degree-1 key; remaining loops start at the lowest unconsumed key. Canonicalise each open chain by choosing the lexicographically smaller endpoint-first key sequence; canonicalise cycles by minimum cyclic key rotation and lexicographically smaller direction. This makes paths and their digest independent of allocation order or view.

05 Ambiguous saddle configurations

Cases 5 and 10 have four crossed cell edges. A fixed diagonal can disconnect a contour incorrectly even if every local endpoint is geometrically sound. Use the **asymptotic decider** for the bilinear interpolant; the determinant is evaluated from unrounded, threshold-relative corner values.



Decider, for g = sample - level

```
Q = gTL*gBR - gTR*gBL

Q > 0 : (Top, Right) and (Bottom, Left)

Q < 0 : (Top, Left) and (Right, Bottom)

Q = 0 : cross at the bilinear saddle (T-B and L-R)
```

Figure 3. Four saddle test cells. Filled nodes are above the level; outlined nodes below it. Lines show the two permitted contour pairings. The exact tie contains two crossing strands, not a single four-way topological junction.

Decision and exact tie

```
q = g_tl * g_br - g_tr * g_bl
if q > 0: [Top--Right, Bottom--Left]
if q < 0: [Top--Left,  Right--Bottom]
if q == 0: [Top--SaddleV--Bottom, Left--SaddleH--Right]
```

For Q = 0, the bilinear saddle lies at `sx = (gTL − gBL)/d`, `sy = (gTL − gTR)/d` in local cell coordinates measured right and down, where `d = gTL − gTR + gBR − gBL`. (Equivalent to sx = −c/d, sy = −b/d in f = a+bx+cy+dxy.) Only the valid interior saddle of an ambiguous cell is used. Two separate virtual keys share the same point: one joins T–B and the other L–R. The contour graph therefore contains two crossing strands, not a degree-4 merged node.

Use the exact Q sign as the topology branch, not an arbitrary center-value test. Define the scale-relative diagnostic **rho = |Q| / max(|gTL·gBR|, |gTR·gBL|)** for ambiguous cells (nonzero denominator). Display “near-degenerate” only when **0 < rho < 0.10**; Q = 0 is instead an exact saddle tie. This is **diagnostic only** and never changes the sign branch or merges contours. A small absolute |Q| alone does not imply small rho or a subpixel gap. Test both ambiguous masks with both signs, the exact tie, complement symmetry, 90° rotation and positive scale invariance of rho.

06 Shared crossing identities and stitching

Build once per level

- **Pass A - crossings:** enumerate H and V grid edges; if endpoint signs differ, calculate a single f64 world point and store it by canonical H/V key.
- **Pass B - cell segments:** use the 16-case table and Q; each ordinary segment joins two canonical crossings. Exact saddle ties add the two distinct virtual saddle vertices.
- **Pass C - adjacency:** build an undirected multigraph with a stable segment ID (cell index, local pair index). Reject missing keys, duplicate segment IDs, nonfinite points and ordinary interior degree other than 2.
- **Pass D - paths:** from all segments, or from the strict row-major cell prefix, emit degree-1 open chains then genuine closed cycles. In a prefix, degree-1 endpoints may lie on the scan frontier; distinct virtual saddle keys keep the tie strands separate.
- **Pass E - validate:** all segment IDs are consumed exactly once; at full completion, ordinary interior endpoints have degree 2, and only exterior-boundary keys may have degree 1. A prefix can expose frontier degree-1 ends, but never phantom joins.

Required topology invariants

Invariant	Observable acceptance condition
Shared edge	An H/V key has bitwise-identical world coordinates when referenced by adjacent cells. No epsilon-based deduplication.
Interior continuity	At full grid completion, every ordinary interior crossing has degree 2, including at closed-loop joins.
Boundary continuity	At full completion degree-1 crossings lie only on the outer boundary; in prefixes they can also lie on the scan frontier. Genuine interior prefix cycles remain closed.
Saddle continuity	For Q = 0, two virtual saddle keys have equal coordinates but remain separate strands. Each has degree 2.
Coverage	Sum of path segment counts equals emitted segment count (splitting tie spokes included); each segment consumed once.
Canonical order	Order by (kind,row,col) with H < V < SaddleV < SaddleH; endpoints interpolate along the canonical edge direction. Equal state yields identical ordered paths and float bits.

Golden single-cell fixtures at threshold 0

Corners (TL,TR,BR,BL)	Mask / Q	Expected pairs
(3, −1, 1, −2)	5 / +1	T–R and B–L
(2, −3, 1, −2)	5 / −4	T–L and R–B
(−3, 1, −2, 2)	10 / +4	T–R and B–L
(−1, 3, −2, 1)	10 / −1	T–L and R–B
(1, −1, 1, −1)	5 / 0	T–saddle–B, L–saddle–R

07 Controller, snapshots and commit policy

Immutable source, small mutable cursor

A completed `Trace` contains the validated grid, three ordered level traces, edge-crossing cache and the deterministic per-cell segment records. The mutable `Cursor` contains {field id, step k, level visibility, inspected level, selected cell, view intent, dirty flag, generation, request token}. No history stack is necessary: `project(trace, k, visibility, selected)` reconstructs the same view from arbitrary seek or reverse navigation.

```
fn navigate(next: State) -> Result<(), Error> {
    if dirty { draw_full(committed_state)?; dirty = false; }
    let projected = project(&trace, &next)?;
    match render(projected) {
        Ok(()) => { committed_state = next; Ok(()) },
        Err(e) => { dirty = true; report(e); Err(e) },
    }
}
```

Transactional expectations

- Compute and validate a new trace for a field switch **or edited threshold** before touching the page. A rejected value (including out-of-order levels or a vertex exactly at level) leaves the previous complete state visible and resets the input to its committed value. Never sort thresholds silently: each value and line/legend style stays in its low, middle or high slot.
- Write the plot geometry with `set_plot_items`, which already replaces plot content transactionally. Additional inspector and status writes are not guaranteed atomic: if any fails, stop progression, retain committed state and mark dirty.
- On the next action, *including a boundary no-op or same-index seek*, repair the old committed projection before processing the requested action. If repair fails, refuse progression and preserve an actionable Retry display control.
- Batch range input to at most one render per animation frame. Every pending request carries **both generation and strictly increasing request token**; callback commits only if both still match. Final change cancels pending input and commits synchronously; cancelled/late callbacks cannot overwrite a newer seek.
- Do not automatically recompute contours during zoom or pan. A view write failure uses the library's own transaction/retry behaviour and must not corrupt controller state.

Readout for each step

The status describes a completed cell only after it was drawn. A normal step says, for example, “Cell (13, 22), high level +0.520: mask 5; Q positive; T–R and B–L; 3 levels visible.” At k=0 with no explicit selection, say “No cell selected”. For an unvisited selected cell say “Not yet evaluated”; for a visited cell report mask, Q, rho and decider. The last step includes per-level open paths, loops and saddle ties. Progress is counted in **cells**, not in edges or contours.

08 Rendering, zoom and reconstruction

Plot geometry and fallback

PlotConfig uses padded linear axes **x = −0.25..(W−0.75)** and **y = −0.25..(H−0.75)**, although sample points remain at integer coordinates 0..W−1 and 0..H−1. The padding keeps valid boundary endpoints off the plot clipping line. Draw three dashed/solid levels using PlotLine; the current plot painter emits SVG <polyline> elements, not <path>. The Phase 0 probe must confirm actual selectors, line styles and three legend entries.

A closed cycle rendered as a repeated-first-point polyline may show a butt-cap seam. Probe at thick line widths and all zoom limits. If a notch is visible, use an actual closed SVG path with Z, as a demo-local overlay **inside the Scene content transform**; never render a nominally closed loop as two visibly disconnected caps. PlotPoints may be used for sample locations. No one-contour-segment-per-graph-edge design is allowed. Detect saddle near-degeneracy in model space through rho, but establish visible separation empirically; do not make a categorical “only visible at 4×” promise.

The stable stage and view compatibility

The stage is one durable outer <svg>. SVG buttons, slider and native selectors occupy its unzoomed control layer. An attached hidden candidate contains only the replaceable Scene plot and its zoomed decorations; it is created with new generation-scoped IDs. External status, controls and event delegation are installed once. A field change preserves view **only** when dimensions and coordinate extents (including axis padding) match; switching to/from the 9×7 coarse preset or 2×2 Saddle lab resets zoom/pan; Ripple ↔ Twin hills preserves view because their sample sizes and axes match.

```
let saved = old_scene.view();
let generation = next_generation();  // even on retry
let candidate = build_hidden_scene(generation, state)?;
if same_dimensions_and_extents { candidate.set_view(saved)?; }
else { candidate.reset_view()?; }
verify_candidate_model_and_geometry(&candidate)?;
swap_only_scene_subtree_atomically()?;
retire_old_scene_and_candidate_handlers();
```

The candidate host remains attached but invisible (visibility:hidden, not display:none) during measurements. On any failure before a successful swap, dispose of it and keep the old working scene. Retrying constructs a **new generation**; old candidates and NodeIds are not reused. Preserve selected cell and keyboard focus by semantic ID, not a stale DOM reference.

Independent continuity requirements

- **Topology:** a canonical digest over levels, keys, ordered path sequences, world f64 bit patterns and open/closed/saddle counts must remain identical for identical field+threshold+step states; view changes never alter it.
- **Rendering:** inspect the browser's actual polyline/path geometry, invert the current CTM and compare independent expected world coordinates, shared-endpoint coincidence and absence of visible closed-loop seams.
- **Tolerance:** do **not** hard-code 1e−6 grid units. Browser SVG geometry readback can be float32. Measure the bound in Phase 0 (anticipated order 1e−5..1e−3 grid units) and freeze an evidence-based tolerance; screen-space comparison is a separate check.

09 Failure modes, lifecycle and budgets

Operation / failure	Required outcome
Invalid dimensions, sample, or level	Typed validation error; old trace, UI choice and visible scene remain committed.
Sample equals threshold	VertexOnLevel; no partial contour or DOM change. Distinct from exact Q=0 saddle.
Excess geometry or invalid arithmetic	GeometryBudgetExceeded { kind, requested, limit } (or typed arithmetic error) before allocation or DOM mutation.
set_plot_items or inspector write fails	Stop navigation, retain committed step; mark dirty if any nontransactional write occurred.
Hidden build or set_view fails	Discard candidate, keep old Scene, old focus and existing controls.
Host scene-subtree swap fails	Old Scene remains active; repair any ancillary partial UI write; a retry gets a NEW generation.
Late seek callback or pagehide	Ignore unless request generation AND request token match. Cancel all frames, detach test hooks and candidate listeners.

Listener ownership and deterministic frame testing

Keep controls and host listeners stable across all reconstructions; candidate listeners have detachable handles and no leaked closure::forget. Range input uses a coalesced request with (trace/rebuild generation, request token). Provide an optional ?manual-clock test mode (or explicit pre-build opt-in), exposing deterministic advance/pending methods only while that mode is active; remove hooks on pagehide, preserving any hook replaced by a later owner.

Checked geometry budgets

Worst-case allowed field is 65×65 samples: **4,096 cells** and three levels. Every ambiguous exact-tie cell can create **four spoke segments**, so a conservative cap is **49,152 raw segments** (4 × 4,096 × 3). The corresponding endpoint-incidence bound is **98,304**. Count stitched path vertices (including closure representation), path objects, plotted points, labels and DOM items separately before allocating or painting. No fixed 12,288-vertex cap: it would reject valid accepted inputs. Enforce typed per-kind caps and checked sizes at every intermediate stage.

Observability

The stable diagnostic footer names the field and levels, selected step/cell, provisional and completed path counts, saddle count, build generation, latest request token, zoom percentage and model digest prefix. Keep test-only introspection opt-in rather than introducing a production global mutable hook.

10 Verification and golden tests

Native tests prove the arithmetic and graph topology; CDP tests prove actual plot-node DOM output, zoom transforms, controls, swaps and failure recovery. Images are only a visual diagnostic, never a substitute for exact key and endpoint checks.

Test suite	Specific assertions
16 cases + symmetry	All masks, complements and 90-degree rotations; stable edge pairing independent of case discovery order.
Saddle fixtures	Both 5/10 masks and Q signs, tie at (0.5,0.5), two virtual degree-2 vertices, no spurious degree-4 node. Test rho < 0.10 separately.
Production Ripple levels	Assert each (row,col), mask and Q sign in the numerical fixture table; +0.101 → +0.102 preserves mask 10 but reverses Q and its pairing; no preset sample within 1e−4.
Coarse preset oracle	9×7 sin(x)sin(y): ambiguous counts 6, 6, 2 at −0.10, +0.10, +0.20; both masks occur at the first two levels. Nonzero-Q topology must match independent oracle.
Interpolation and ordering	Plane x−a has exact straight boundary-to-boundary open isoline; H/V interpolation direction and (kind,row,col) order are deterministic.
Loop / prefix invariants	A closed peak yields a cycle; partially visited fields permit frontier degree-1 ends and already completed closed cycles; no premature join.
Shared keys and coverage	Adjacent cells reference the same crossing; full interior degree 2, boundary degree 1; every segment accounted for once.
Zoom and DOM readback	Test nested pointer-centred wheel zoom, pan and toolbar position at 0.25×, 1×, 4×; inverse CTM, actual <polyline>/<path> types and measured tolerance.
Reconstruction ×25	No changed digest at identical logical states, altered geometry, stale SVG, duplicate listener, lost focus or changed view.
Input races and faults	Delayed/cancelled token, failed plot/control write, hidden build and subtree swap; retry uses a fresh generation.
Accessibility and budget	Fixed-size stage controls; keyboard-only inspection, live region and legend; checked 49,152 segment/98,304 incidence limits.

Independent reference

A separate native small-grid oracle must enumerate crossings and edge pairings without importing the production case table or stitcher. Check golden manifests for four nonzero saddles, the tie, an open vertical cut, a closed cycle, the two 25×19 fields and the 9×7 coarse preset, including per-level mask/Q checks. Compare semantic key paths and f64 bits rather than SVG attribute ordering.

Because local macOS CDP execution has previously been unstable, run native and WASM tests locally and require full browser geometry, clipping, seam and accessibility tests on supported CI runners; record any browser gate not executed.

11 Delivery phases and acceptance gates

Phase	Deliverables	Gate
0 - API and DOM probe	Probe polyline seam, padded clipping, float32 readback, legend, pointer-centred wheel zoom, pan, toolbar position and Scene view restoration in nested SVG.	Freeze actual DOM selectors, empirically justified tolerances and seam/clipping fallback before feature work.
1 - Pure solver	Grid validation, ascending levels, 16 cases, Q and rho, exact tie, canonical cache, stitching, coarse fixture and independent oracle.	All mask and topology invariants pass without browser.
2 - Static visualisation	Ripple default, Twin hills, Coarse saddle waves and Saddle lab; slot-linked legend/style, accurate inspector and keyboard controls.	Rendering corresponds to validated world paths and all thresholds are visibly distinct.
3 - Navigation	Cell-by-cell trace, previous/next/seek, field or numeric threshold change, dirty-state retry and error reporting.	Random seek/reverse and prefix-cycle tests; request-token races and dirty repair.
4 - Rebuild / zoom	Stable outer SVG, transactional Scene-only swap, compatible view/focus, fresh generations and CDP geometry checks.	25 rebuilds and threshold reversal: identical-state digest, no broken paths or extra handlers.
5 - Documentation / CI	Demo page, gallery/source registration, test docs, benchmark budgets, short README and screenshot.	All acceptance checks green on supported runners and source gallery works. The V3 fixture matrix and saddle rho rule are documented.

Proposed code placement

```
demo-app/src/marching_squares/{mod,field,solver,stitch}.rs
demo-app/src/marching_squares/{projection,controller,page}.rs
demo-app/src/marching_squares/{inspector,layout,dom}.rs
demo/panels/<ordered-group>/marching-squares.html
cdp-integration-test/tests/cdp/marching_squares.rs
docs/demo/marching-squares.md
```

Register one `demo_gallery!` entry and the source-file list in `demo-app/src/lib.rs`; use the existing demo grouping and panel naming scheme. Add the panel fragment so the demo server catalogue validator agrees. Native tests stay with the solver/projection; browser tests are collected under the existing CDP suite. If any path or callable API differs on the implementation branch, Phase 0 documents the adjustment.

Definition of done

- The default Ripple levels and the 9×7 coarse preset reproduce their golden Q/mask fixtures; neither relies on an exactly-zero-Q production sample; inspector is keyboard-accessible.
- Every nondegenerate contour has the expected open/closed topology and shared-edge continuity; 5/10 saddles are resolved by the determinant, including the exact crossing fixture.
- At matching logical states after 25 rebuilds, the digest, contour counts and model-world geometry agree; DOM geometry, selected focus and single-bound listeners pass independent checks.
- Write failures do not advance the committed step or silently damage the scene; the next action repairs or reports the fault.
- The demo requires no new public svg-dom-graph algorithm API and is integrated into the gallery, documentation and CI.

12 Source anchors and engineering notes

The design deliberately follows the current workspace model: pure Rust performs the arithmetic; Scene and PlotNode draw caller-provided results; gallery panels are registered independently; controls and browser testing follow existing examples. Verify against the implementation branch during Phase 0 because source APIs can change.

Repository anchors (accessed 4 November 2026)

Reference	Use in this design
github.com/ChrisWhealy/svg-dom-graph/blob/main/docs/library/plot-nodes.md	PlotConfig, PlotLine, PlotPoints, set_plot_items, description, click semantics and clipping.
.../docs/library/toolbar-and-zoom.md	0.25–4.0 view range; view/set_view and shared content transform.
.../docs/library/view-updates.md	Zoom is a transform-only update; no geometry or connector rerouting.
.../docs/demo/n-queens.md	Pure snapshot projection, coalesced seek, dirty-state repair and controller tests.
.../docs/demo/red-black-tree.md	Hidden attached scene, atomically swapped DOM, keyboard focus restoration.
.../docs/demo/fft.md	Multiple independent views, full off-screen rebuild and preserved zoom.
.../docs/demo/demo-internals.md	Gallery fragment naming, source registration and build integration.

Mathematical reference

Nielson, G. M. and Hamann, B. (1991), *The Asymptotic Decider: Resolving the Ambiguity in Marching Cubes*, IEEE Visualization '91, DOI 10.1109/VISUAL.1991.175782. The method motivates the bilinear saddle decision used for the ambiguous 2D cell cases. See also Athawale et al. (2018), *Probabilistic Asymptotic Decider for Topological Ambiguity Resolution in Level-Set Extraction for Uncertain 2D Data*, IEEE Transactions on Visualization and Computer Graphics, for the distinction between the midpoint and saddle-point deciders.

Explicit design decisions

- The **source field** is immutable per trace; only the projection and view change. Thresholds remain strictly ascending and style identities stay attached to low/middle/high slots.
- The **saddle tie** is a genuine pair of intersecting strands with separate virtual vertex identities.
- A sampled vertex **exactly on a level** is rejected in V3, rather than silently producing undefined plateau topology.
- The **zoom test** compares geometry in logical coordinates after inverse transforms, not raw screenshot pixels.
- A **rebuild** is an atomic replacement of a complete candidate and never a partial teardown/recreate of the visible scene.

Review readiness: *V3 incorporates the V2 review plus normalized-Q diagnostics, coarse preset and fixed threshold-slot identities, subject to Phase 0 nested-view, readback, seam and clipping probes. No implementation or repository modifications are claimed.*

Appendix A Phase 0 integration protocol

Capture results on the actual implementation branch; do not substitute assumptions about svg-dom-graph's API or a browser's SVG geometry precision. Phase 1 is blocked until the following probes have a recorded verdict.

Probe	Evidence required / resolution
DOM primitive	Inspect rendered PlotLine elements and readback API. Current painter emits <polyline>. Freeze selectors for both polyline and a potential closed-path fallback.
Coordinate precision	At known f64 world points, compare inverse-CTM SVG geometry for 1×, 0.25× and 4×. Establish an absolute grid-unit error bound, not a guessed 1e−6.
Clipping/padding	Render a known half-plane whose contour meets the outer top and bottom sample boundaries. Confirm padded axes do not drop endpoints.
Closed-loop join	Draw a thick closed contour, measure/inspect the first-last seam at 1× and both zoom limits. If visible, implement a true closed Z path inside Scene transform and retest.
Three-entry legend	Render solid, dashed and dotted entries with labels simultaneously; ensure each entry matches the actual plotted line and accessible description.
Stage controls	At 0.25× and 4×, verify fixed-size buttons/range/selectors in stage SVG and toolbar position outside clipped plot; keyboard focus survives a Scene-only swap.
Nested pointer zoom/pan	With Scene <svg> nested in stage <svg>, wheel zoom about a chosen off-centre pointer location; verify anchor under pointer remains fixed in stage coordinates, pan movement follows gesture, and toolbar anchors remain stable before/after rebuild.
Hidden candidate	Probe attached visibility:hidden layout, setting/restoring view, atomic subtree swap and teardown. A failed build must leave the former visible Scene working.
Clock and lifecycle	In ?manual-clock mode, queue multiple input updates, cancel, create a newer token, release the old callback and prove it cannot commit; pagehide frees hooks.

Probe report format

Record: browser/version; crate revision; tested plot type; world→SVG mapping; measured max geometry deviation and chosen tolerance; clipping result; closed-seam result; legend result; stage control result; pointer-anchored wheel/pan/toolbar result before and after swap; hidden-swap failure behaviour; and exact selector names. If a probe fails, document the permitted demo-local fallback without changing the pure Rust solver contract.

Appendix B End-to-end continuity regression

This is an additional acceptance sequence, exercised with the Ripple middle-level saddle at cell (14,12) and an explicit inspected cell. Run the whole chain without reloading the page, then repeat the reconstruction operation 25 times at a nonidentity view.

```
A. Start at Ripple defaults; capture model digest D0 and DOM geometry.
B. Edit middle threshold +0.101 to +0.102; case 10 stays,
   Q flips from -4.83e-5 to +4.31e-5; pairing flips; capture D1.
C. Zoom about off-centre pointer to 0.25x, 4x; pan; D1 fixed.
D. Rebuild Scene; D1 and actual world geometry must agree.
E. Restore +0.101; recovered model digest must equal D0.
F. Rebuild again; preserve focus, view and geometry.
G. Reset zoom; D0 must still be unchanged.
```

At every checkpoint

- Compare model topology and digest only against a checkpoint with the **same** field, threshold configuration and processed-cell prefix; the +0.101 → +0.102 test must change the inspected case-10 pairing, while reverse editing restores D0.
- Verify shared H/V crossings agree exactly in the model; full-grid ordinary interior degree is 2; prefix frontier free ends and genuine already closed cycles are classified correctly.
- Verify actual browser-drawn polyline/path endpoints in world coordinates through inverse CTM. Confirm boundary endpoints are not clipped and closed contours have no visible notch.
- Verify independent per-level open, closed and exact-saddle counts and confirm every raw segment is used once; no surplus, missing or stale SVG elements.
- Check focus by semantic selection, pointer-centred zoom and pan, toolbar anchoring, stable control size, listener cardinality and generation + request-token guards after reconstruction and after a late callback.
- Inject a failed candidate swap and a failed plot/inspector write. The old committed state remains usable or repairs on the next action (even a boundary no-op). Rebuild retry receives a fresh generation.

Pass criterion

The 25-cycle run and the full threshold→zoom→rebuild→reverse→rebuild→reset sequence pass both **model/topology** and **real DOM/visual geometry** checks. Equal digests alone are insufficient: transformed-but-disconnected artwork or a butt-cap seam must fail the visual gate. The regression is deterministic under the optional manual-frame mode.